Day - 1 (Download And Install Docker On Window & CLI)

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★ Installing Docker On Window:

You can install it from google: [Docker Desktop]

★ Open The Terminal:

- Write first to list all the commands: **docker --help**

```
PS C:\Users\ayush> docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Log in to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  builder     Manage builds
  buildx*     Docker Buildx
  compose*    Docker Compose
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container
  dev*        Docker Dev Environments
  extension*  Manages Docker extensions
  feedback*   Provide feedback, right in your terminal!
```

★ Installing Docker On Linux:

- **Define Version Of Ubuntu:** cat /etc/os-release

```
ubuntu@ip-172-31-51-74:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

- **Version Check:** lsb_release -cs

```
ubuntu@ip-172-31-51-74:~$ lsb_release -cs
No LSB modules are available.
noble
```

- **Verify Processor Architecture:** uname -m

```
ubuntu@ip-172-31-51-74:~$ uname -m
x86_64
```

★ Guide: https://docs.docker.com/engine/install/ubuntu/

- Different Architecture Installation Process Is Different.

★ Check the machine (Ubuntu):

Commands Are:

- To Check the Machine Name: **cat /etc/os-release**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04 LTS"
Files Ubuntu"
VERSION_ID="24.04"
VERSION="24.04 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-
cy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

- To check the version: **lsb_release -cs**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ lsb_release -cs
No LSB modules are available.
noble
```

- To check the processor architecture: **usnmae -m**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ uname -m
x86_64
```

★ Install Docker In CLI:

- Update Information About Packages: **sudo apt-get update**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo apt-get update
[sudo] password for ayush-sharma:
Warning: The unit file, source configuration file or drop-ins of apt-news.servic
e changed on disk. Run 'systemctl daemon-reload' to reload units.
Warning: The unit file, source configuration file or drop-ins of esm-cache.servi
ce changed on disk. Run 'systemctl daemon-reload' to reload units.
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [348
 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages
[323 kB]
Fetched 797 kB in 4s (226 kB/s)
Reading package lists... Done
```

- Install Some Packages: **sudo apt-get install ca-certificates curl**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo apt-get install ca-certificates cur
l
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 227 kB of archives.
After this operation, 532 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.
5.0-2ubuntu10.2 [227 kB]
Fetched 227 kB in 3s (67.2 kB/s)
Selecting previously unselected package curl.
(Reading database ... 147884 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.2_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.2) ...
Setting up curl (8.5.0-2ubuntu10.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

- **Other Commands:**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo install -m 0755 -d /etc/apt/keyring
s
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo curl -fsSL https://download.docker.
com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo chmod a+r /etc/apt/keyrings/docker.
asc
ayush-sharma@ayush-sharma-VirtualBox:~$ echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc
] https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

- **Update Information About Packages:**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo apt-get update
Warning: The unit file, source configuration file or drop-ins of apt-news.servic
e changed on disk. Run 'systemctl daemon-reload' to reload units.
Warning: The unit file, source configuration file or drop-ins of esm-cache.servi
ce changed on disk. Run 'systemctl daemon-reload' to reload units.
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.8 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:3 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [12.4
 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [294
kB]
Fetched 608 kB in 2s (325 kB/s)
Reading package lists... Done
```

- **Install Docker Packages:** (Docker Community Engine)

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo apt-get install docker-ce docker-ce
-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras git git-man liberror-perl libslirp0 pigz
  slirp4netns
Suggested packages:
  aufs-tools cgroupfs-mount | cgroup-lite git-daemon-run | git-daemon-sysvinit
  git-doc git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli
  docker-ce-rootless-extras docker-compose-plugin git git-man liberror-perl
  libslirp0 pigz slirp4netns
0 upgraded, 12 newly installed, 0 to remove and 2 not upgraded.
Need to get 127 MB of archives.
After this operation, 461 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io
```

- **Running Docker Image:** sudo docker run hello-world

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
c1ec31eb5944: Pull complete
Digest: sha256:53cc4d415d839c98be39331c948609b659ed725170ad2ca8eb36951288f81b75
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.
```

- **Check Images:** sudo docker images

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED         SIZE
hello-world   latest    d2c94e258dcb   15 months ago   13.3kB
```

- Run Another Container: **sudo docker run ubuntu**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo docker run ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
31e907dcc94a: Pull complete
Digest: sha256:8a37d68f4f73ebf3d4efafbcf66379bf3728902a8038616808f04e34a9ab63ee
Status: Downloaded newer image for ubuntu:latest
```

- **(History Of Containers) Docker that exited:** sudo docker ps -a
  sudo docker container ls -a

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo docker ps -a
CONTAINER ID   IMAGE         COMMAND       CREATED          STATUS
        PORTS     NAMES
f49f6a8026a2   ubuntu        "/bin/bash"   3 minutes ago    Exited (0) 3 minutes
 ago              serene_cartwright
aea3cd02ad3f   hello-world   "/hello"      25 minutes ago   Exited (0) 25 minute
s ago             wonderful_fermi
```

- Log In Ubuntu Container: sudo docker run -it ubuntu

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo docker run -it ubuntu
root@7e249e64aa24:/#
```

Now docker in running state. You can check it using (**sudo docker ps**)

- List All Files In Ubuntu Container: **ls**

```
ayush-sharma@ayush-sharma-VirtualBox:~$ sudo docker run -it ubuntu
root@7e249e64aa24:/# ls
bin   dev  home  lib64  mnt  proc  run   srv  tmp  var
boot  etc  lib   media  opt  root  sbin  sys  usr
```

Inside Bin Files Use Command (ls bin)

- Echo Command:

```
root@7e249e64aa24:/# echo "Hello From Ubuntu Container"
Hello From Ubuntu Container
```

- **hostname -i**: It show Ip Address

```
root@7e249e64aa24:/# hostname
7e249e64aa24
root@7e249e64aa24:/# hostname -i
172.17.0.2
```

★ Running BusyBox Container:

- Pull Docker Image By Run Command:

```
PS C:\Users\ayush> docker run busybox
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
a46fbb00284b: Pull complete
Digest: sha256:768e5c6f5cb6db0794eec98dc7a967f406317 46c32232b78a3105fb946f3ab83
Status: Downloaded newer image for busybox:latest
```

Images Have Specific Tag

- **List Some Commands In BusyBox Image:**

```
PS C:\Users\ayush> docker run -it busybox
/ #
/ #
/ #
/ #
```

BusyBox is not a fully featured of linux and it is set of utilities of linux operating system.

```
/ # ls
bin    dev    etc    home   lib    lib64  proc   root   sys    tmp    usr    var
/ # ls bin
[                  dos2unix           ifconfig           mknod              rmmod              test
[[                 dpkg               ifdown             mkpasswd           route              tftp
acpid              dpkg-deb           ifenslave          mkswap             rpm                tftpd
add-shell          du                 ifplugd            mktemp             rpm2cpio           time
addgroup           dumpkmap           ifup               modinfo            rtcwake            timeout
adduser            dumpleases         inetd              modprobe           run-init           top
adjtimex           echo               init               more               run-parts          touch
ar                 ed                 insmod             mount              runlevel           tr
arch               egrep              install            mountpoint         runsv              traceroute
arp                eject              ionice             mpstat             runsvdir           traceroute6
```

```
/ # echo "Hello From BusyBoc"
Hello From BusyBoc
/ # uptime
 05:17:04 up 18 min,  0 users,  load average: 0.01, 0.03, 0.00
/ #
```

★ Compare Alpine And Busybox:

```
PS C:\Users\ayush> docker run busybox
PS C:\Users\ayush> docker run -it busybox
/ # ls
bin    dev    etc    home   lib    lib64  proc   root   sys    tmp    usr    var
/ # ls /bin
[                  expand             link               poweroff           sync
[[                 expr               linux32            powertop           sysctl
acpid              factor             linux64            printenv           syslogd
add-shell          fakeidentd         linuxrc            printf             tac
addgroup           fallocate          ln                 ps                 tail
adduser            false              loadfont           pscan              tar
adjtimex           fatattr            loadkmap           pstree             taskset
ar                 fbset              logger             pwd                tc
arch               fbsplash           login              pwdx               tcpsvd
arp                fdflush            logname            raidautorun        tee
arping             fdformat           logread            rdate              telnet
ascii              fdisk              losetup            rdev               telnetd
ash                fgconsole          lpd                readahead          test
awk                fgrep              lpq                readlink           tftp
base32             find               lpr                readprofile        tftpd
base64             findfs             ls                 realpath           time
basename           flock              lsattr             reboot             timeout
bc                 fold               lsmod              reformime          top
beep               free               lsof               remove-shell       touch
blkdiscard         freeramdisk        lspci              renice             tr
blkid              fsck               lsscsi             reset              traceroute
blockdev           fsck.minix         lsusb              resize             traceroute6
bootchartd         fsfreeze           lzcat              resume             tree
brctl              fstrim             lzma               rev                true
bunzip2            fsync              lzop               rm                 truncate
busybox            ftpd               makedevs           rmdir              ts
bzcat              ftpget             makemime           rmmod              tsort
bzip2              ftpput             man                route              tty
cal                fuser              md5sum             rpm                ttysize
cat                getconf            mdev               rpm2cpio           tunctl
chat               getfattr           mesg               rtcwake            ubiattach
chattr             getopt             microcom           run-init           ubidetach
chgrp              getty              mim                run-parts          ubimkvol
chmod              grep               mkdir              runlevel           ubirename
chown              groups             mkdosfs            runsv              ubirmvol
chpasswd           gunzip             mke2fs             runsvdir           ubirsvol
chpst              gzip               mkfifo             rx                 ubiupdatevol
chroot             halt               mkfs.ext2          script             udhcpc
chrt               hd                 mkfs.minix         scriptreplay       udhcpc6
chvt               hdparm             mkfs.vfat          sed                udhcpd
cksum              head               mknod              seedrng            udpsvd
clear              hexdump            mkpasswd           sendmail           uevent
cmp                hexedit            mkswap             seq                umount
comm               hostid             mktemp             setarch            uname
conspy             hostname           modinfo            setconsole         unexpand
cp                 httpd              modprobe           setfattr           uniq
```

```
PS C:\Users\ayush> docker run alpine
PS C:\Users\ayush> docker run -it alpine
/ # ls
bin    etc    lib    mnt    proc   run    srv    tmp    var
dev    home   media  opt    root   sbin   sys    usr
/ # ls /bin
arch           dmesg          hostname       makemime       pipe_progress  stat
ash            dnsdomainname  ionice         mkdir          printenv       stty
base64         dumpkmap       iostat         mknod          ps             su
bbconfig       echo           ipcalc         mktemp         pwd            sync
busybox        egrep          kbd_mode       more           reformime      tar
cat            false          kill           mount          rev            touch
chattr         fatattr        link           mountpoint     rm             true
chgrp          fdflush        linux32        mpstat         rmdir          umount
chmod          fgrep          linux64        mv             run-parts      uname
chown          fsync          ln             netstat        sed            usleep
cp             getopt         login          nice           setpriv        watch
date           grep           ls             pidof          setserial      zcat
dd             gunzip         lsattr         ping           sh
df             gzip           lzop           ping6          sleep
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.3
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # ls etc
alpine-release   inittab          network          protocols        ssl1.1
apk              issue            nsswitch.conf    resolv.conf      sysctl.conf
busybox-paths.d  logrotate.d      opt              secfixes.d       sysctl.d
crontabs         modprobe.d       os-release       securetty        udhcpc
fstab            modules          passwd           services
group            modules-load.d   periodic         shadow
hostname         motd             profile          shells
hosts            mtab             profile.d        ssl
/ # ls -la bin
total 800
drwxr-xr-x    2 root     root          4096 Sep  6 11:34 .
drwxr-xr-x    1 root     root          4096 Oct  5 05:36 ..
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 arch -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ash -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 base64 -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 bbconfig -> /bin/busybox
-rwxr-xr-x    1 root     root        808712 Jun 10 07:11 busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 cat -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 chattr -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 chgrp -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 chmod -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 chown -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 cp -> /bin/busybox
```

```
cpio            hush            more            setfont         unix2dos
crc32           hwclock         mount           setkeycodes     unlink
crond           i2cdetect       mountpoint      setlogcons      unlzma
crontab         i2cdump         mpstat          setpriv         unshare
cryptpw         i2cget          mt              setserial       unxz
cttyhack        i2cset          mv              setsid          unzip
cut             i2ctransfer     nameif          setuidgid       uptime
date            id              nanddump        sh              users
dc              ifconfig        nandwrite       sha1sum         usleep
dd              ifdown          nbd-client      sha256sum       uudecode
deallocvt       ifenslave       nc              sha3sum         uuencode
delgroup        ifplugd         netstat         sha512sum       vconfig
deluser         ifup            nice            showkey         vi
depmod          inetd           nl              shred           vlock
devmem          init            nmeter          shuf            volname
df              insmod          nohup           slattach        w
dhcprelay       install         nologin         sleep           wall
diff            ionice          nproc           smemcap         watch
dirname         iostat          nsenter         softlimit       watchdog
dmesg           ip              nslookup        sort            wc
dnsd            ipaddr          ntpd            split           wget
dnsdomainname   ipcalc          od              ssl_client      which
dos2unix        ipcrm           openvt          start-stop-daemon who
dpkg            ipcs            partprobe       stat            whoami
dpkg-deb        iplink          passwd          strings         whois
du              ipneigh         paste           stty            xargs
dumpkmap        iproute         patch           su              xxd
dumpleases      iprule          pgrep           sulogin         xz
echo            iptunnel        pidof           sum             xzcat
ed              kbd_mode        ping            sv              yes
egrep           kill            ping6           svc             zcat
eject           killall         pipe_progress   svlogd          zcip
env             killall5        pivot_root      svok
envdir          klogd           pkill           swapoff
envuidgid       last            pmap            swapon
ether-wake      less            popmaildir      switch_root
/ # cat /etc/os-release
cat: can't open '/etc/os-release': No such file or directory
/ # ls etc
group           hosts           mtab            nsswitch.conf   resolv.conf
hostname        localtime       network         passwd          shadow
/ # ls -la bin
total 408284
drwxr-xr-x    2 root     root         12288 Sep 26 21:31 .
drwxr-xr-x    1 root     root          4096 Oct  5 05:36 ..
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 [
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 [[
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 acpid
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 add-shell
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 addgroup
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 adduser
```

```
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 date -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 dd -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 df -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 dmesg -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 dnsdomainname -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 dumpkmap -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 echo -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 egrep -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 false -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 fatattr -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 fdflush -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 fgrep -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 fsync -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 getopt -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 grep -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 gunzip -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 gzip -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 hostname -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ionice -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 iostat -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ipcalc -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 kbd_mode -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 kill -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 link -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 linux32 -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 linux64 -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ln -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 login -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ls -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 lsattr -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 lzop -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 makemime -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 mkdir -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 mknod -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 mktemp -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 more -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 mount -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 mountpoint -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 mpstat -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 mv -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 netstat -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 nice -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 pidof -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ping -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ping6 -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 pipe_progress -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 printenv -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 ps -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 pwd -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 reformime -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 rev -> /bin/busybox
```

```
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sha256sum
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sha3sum
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sha512sum
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 showkey
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 shred
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 shuf
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 slattach
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sleep
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 smemcap
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 softlimit
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sort
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 split
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 ssl_client
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 start-stop-daemon
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 stat
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 strings
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 stty
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 su
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sulogin
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sum
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sv
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 svc
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 svlogd
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 svok
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 swapoff
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 swapon
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 switch_root
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sync
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 sysctl
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 syslogd
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tac
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tail
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tar
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 taskset
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tc
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tcpsvd
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tee
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 telnet
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 telnetd
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 test
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tftp
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tftpd
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 time
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 timeout
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 top
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 touch
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tr
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 traceroute
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 traceroute6
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 tree
-rwxr-xr-x  405 root     root       1029688 Sep 26 21:31 true
```

```
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 rm -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 rmdir -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 run-parts -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 sed -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 setpriv -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 setserial -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 sh -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 sleep -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 stat -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 stty -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 su -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 sync -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 tar -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 touch -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 true -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 umount -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 uname -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 usleep -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 watch -> /bin/busybox
lrwxrwxrwx    1 root     root            12 Sep  6 11:34 zcat -> /bin/busybox
/ # busybox
BusyBox v1.36.1 (2024-06-10 07:11:47 UTC) multi-call binary.
BusyBox is copyrighted by many authors between 1998-2015.
Licensed under GPLv2. See source distribution for detailed
copyright notices.

Usage: busybox [function [arguments]...]
   or: busybox --list[-full]
   or: busybox --install [-s] [DIR]
   or: function [arguments]...

        BusyBox is a multi-call binary that combines many common Unix
        utilities into a single executable.  Most people will create a
        link to busybox for each function they wish to use and BusyBox
        will act like whatever it was invoked as.

Currently defined functions:
        [, [[, acpid, add-shell, addgroup, adduser, adjtimex, arch, arp, arping, ash, awk, base64,
        basename, bbconfig, bc, beep, blkdiscard, blkid, blockdev, brctl, bunzip2, bzcat, bzip2, cal,
        cat, chattr, chgrp, chmod, chown, chpasswd, chroot, chvt, cksum, clear, cmp, comm, cp, cpio,
        crond, crontab, cryptpw, cut, date, dc, dd, deallocvt, delgroup, deluser, depmod, df, diff,
        dirname, dmesg, dnsdomainname, dos2unix, du, dumpkmap, echo, egrep, eject, env, ether-wake,
        expand, expr, factor, fallocate, false, fatattr, fbset, fbsplash, fdflush, fdisk, fgrep,
        find, findfs, flock, fold, free, fsck, fstrim, fsync, fuser, getopt, getty, grep, groups,
        gunzip, gzip, halt, hd, head, hexdump, hostid, hostname, hwclock, id, ifconfig, ifdown,
        ifenslave, ifup, init, inotifyd, insmod, install, ionice, iostat, ip, ipaddr, ipcalc, ipcrm,
        ipcs, iplink, ipneigh, iproute, iprule, iptunnel, kbd_mode, kill, killall, killall5, klogd,
        last, less, link, linux32, linux64, ln, loadfont, loadkmap, logger, login, logread, losetup,
        ls, lsattr, lsmod, lsof, lsusb, lzcat, lzma, lzop, lzopcat, makemime, md5sum, mdev, mesg,
        microcom, mkdir, mkdosfs, mkfifo, mkfs.vfat, mknod, mkpasswd, mkswap, mktemp, modinfo,
        modprobe, more, mount, mountpoint, mpstat, mv, nameif, nanddump, nandwrite, nbd-client, nc,
```

```
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 ubiattach
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 ubidetach
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 ubimkvol
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 ubirename
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 ubirmvol
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 ubirsvol
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 ubiupdatevol
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 udhcpc
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 udhcpc6
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 udhcpd
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 udpsvd
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 uevent
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 umount
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 uname
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 unexpand
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 uniq
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 unix2dos
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 unlink
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 unlzma
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 unshare
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 unxz
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 unzip
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 uptime
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 users
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 usleep
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 uudecode
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 uuencode
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 vconfig
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 vi
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 vlock
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 volname
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 w
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 wall
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 watch
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 watchdog
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 wc
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 wget
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 which
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 who
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 whoami
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 whois
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 xargs
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 xxd
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 xz
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 xzcat
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 yes
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 zcat
-rwxr-xr-x  405 root      root       1029688 Sep 26 21:31 zcip
/ # apk
sh: apk: not found
/ #
```

```
        readahead, readlink, realpath, reboot, reformime, remove-shell, renice, reset, resize, rev,
        rfkill, rm, rmdir, rmmod, route, run-parts, sed, sendmail, seq, setconsole, setfont,
        setkeycodes, setlogcons, setpriv, setserial, setsid, sh, sha1sum, sha256sum, sha3sum,
        sha512sum, showkey, shred, shuf, slattach, sleep, sort, split, stat, strings, stty, su, sum,
        swapoff, swapon, switch_root, sync, sysctl, syslogd, tac, tail, tar, tee, test, time,
        timeout, top, touch, tr, traceroute, traceroute6, tree, true, truncate, tty, ttysize, tunctl,
        udhcpc, udhcpc6, umount, uname, unexpand, uniq, unix2dos, unlink, unlzma, unlzop, unshare,
        unxz, unzip, uptime, usleep, uudecode, uuencode, vconfig, vi, vlock, volname, watch,
        watchdog, wc, wget, which, who, whoami, whois, xargs, xxd, xzcat, yes, zcat, zcip
/ # apk
apk-tools 2.14.4, compiled for x86_64.

usage: apk [<OPTIONS>...] COMMAND [<ARGUMENTS>...]

Package installation and removal:
  add        Add packages to WORLD and commit changes
  del        Remove packages from WORLD and commit changes

System maintenance:
  fix        Fix, reinstall or upgrade packages without modifying WORLD
  update     Update repository indexes
  upgrade    Install upgrades available from repositories
  cache      Manage the local package cache

Querying package information:
  info       Give detailed information about packages or repositories
  list       List packages matching a pattern or other criteria
  dot        Render dependencies as graphviz graphs
  policy     Show repository policy for packages
  search     Search for packages by name or description

Repository maintenance:
  index      Create repository index file from packages
  fetch      Download packages from repositories to a local directory
  manifest   Show checksums of package contents
  verify     Verify package integrity and signature

Miscellaneous:
  audit      Audit system for changes
  stats      Show statistics about repositories and installations
  version    Compare package versions or perform tests on version strings

This apk has coffee making abilities.
For more information: man 8 apk
/ # busybox ping google.com
PING google.com (142.250.194.238): 56 data bytes
64 bytes from 142.250.194.238: seq=0 ttl=63 time=19.488 ms
64 bytes from 142.250.194.238: seq=1 ttl=63 time=19.406 ms
64 bytes from 142.250.194.238: seq=2 ttl=63 time=19.376 ms
64 bytes from 142.250.194.238: seq=3 ttl=63 time=18.224 ms
64 bytes from 142.250.194.238: seq=4 ttl=63 time=17.854 ms
```